

BUSINESS ANALYTICS PROJECT

Business Analytics of Branded Electronics Sales

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1. Introduction

Business Analytics is the process of collecting, analyzing and interpreting data to understand business performance and support better decision-making. It helps businesses identify patterns, trends and relationships in their data.

Data is important for businesses because it provides useful information about products, customers, sales, pricing, marketing and business performance. By analyzing data, businesses can make decisions based on actual information rather than assumptions.

In this project, the selected business area is Branded Electronics Sales. The dataset contains information about different electronics brands, products, cities, customers, advertising methods, price, quantity, discount, revenue, marketing spend, ratings and demand.

This business area was selected because the electronics market is highly competitive and businesses need to understand which brands and products perform better.

Business Analytics can help this business by analyzing sales performance, customer behaviour, product demand, marketing effectiveness and revenue trends. The analysis can support better sales planning, marketing decisions, product management and future business planning.

2. Problem Statement

The selected branded electronics business needs to understand its sales performance and customer behaviour effectively.

The available dataset contains information about different brands, products, cities, customer types, advertising methods, prices, discounts, revenue, marketing spend, ratings and demand. The project aims to analyze this data to identify important sales patterns, product performance, customer trends, relationships between business variables and future revenue trends.

The analysis will help in making better decisions related to product selection, sales planning, marketing strategy, customer management and revenue forecasting.

3. Objectives

1. To collect and understand the branded electronics sales dataset.
2. To perform data preprocessing and Exploratory Data Analysis.
3. To analyze brand, product, customer and sales performance.
4. To identify relationships between important business variables.
5. To study the effect of marketing and other factors on revenue.
6. To forecast future sales/revenue trends.
7. To perform statistical analysis and hypothesis testing.
8. To provide useful business insights and recommendations.

4. Business Scenario / Domain Description

Business Name: Branded Electronics Retail Business

Type of Business: Retail business selling branded electronic products.

Products Offered

- Smartphones
- Laptops
- TVs
- Headphones
- Tablets

Brands

- Apple
- Samsung
- Sony
- OnePlus
- Dell
- HP

Target Customers:

The business targets new customers and returning customers interested in purchasing branded electronic products.

Business Operations:

The business sells different branded electronics to customers in different cities. It uses different advertising methods such as Social Media, Email and Online Ads to promote products. The business records information about product prices, quantity sold, discounts, revenue, marketing expenditure, customer ratings and demand.

Important Business Problems

The business needs to understand:

- Which brand generates the highest revenue?
- Which product performs better?
- Which cities have better sales?
- How does marketing spend affect revenue?
- Which customer group is more active?
- Which advertising method performs better?
- How can future revenue be predicted?

5. Dataset Description

Dataset Information:

Information	Details
Dataset Name	Branded Electronics Business Analytics Dataset
File Name	Branded_Electronics_Business_Analytics_Project_Dataset.csv
Number of records	180 records
Number of columns	17 columns
Business Domain	Branded Electronics Retail Sales

Description of Columns:

Column	Description	Data Type	Variable Type
Order_id	Unique Order Number	Object	Categorical
Date	Order Date	Date/Time	Date/Time
Brand	Electronic Brand	Object	Categorical
Product	Product Type	Object	Categorical
City	Customer city	Object	Categorical
Customer_Type	New or returning customer	Object	Categorical
AB_Group	A/B testing group	Object	Categorical
Advertising_Type	Advertisement type	Object	Categorical
Price	Product price	Float	Numerical
Quantity	Quantity purchased	Integer	Numerical
Discount_Percent	Discount given	Integer	Numerical
Revenue	Total sales revenue	Float	Numerical
Marketing_Spend	Marketing cost	Float	Numerical
Rating	Customer rating	Float	Numerical
Demand	Product demand	Integer	Numerical
Converted	Customer conversion status	integer	Binary
Sales_Units	Units sold	Integer	Numerical

Numerical Variables

Price, Quantity, Discount_Percent, Revenue, Marketing_Spend, Rating, Demand, Converted and Sales_Units.

Categorical Variables

Brand, Product, City, Customer_Type, AB_Group and Advertising_Type.

Date-Time Variable

Date.

6. Data Collection

The data used for this project is a business-generated academic dataset created for analyzing branded electronics sales.

The dataset was prepared to represent realistic electronics business transactions and contains information related to products, brands, customers, marketing and sales.

Data Collection Method

The dataset was created and stored in CSV format and imported into Python using the Pandas library for analysis.

Number of Records

180 records

Time Period

January 2025 to December 2025

Variables Collected

The dataset contains information about:

- Orders
- Date
- Brand
- Product
- City
- Customer type
- Advertising type
- Price
- Quantity
- Discount
- Revenue
- Marketing spend
- Rating
- Demand
- Conversion
- Sales units

7. Data Preprocessing

Data preprocessing is the process of cleaning and preparing the Branded Electronics Sales dataset before performing analysis.

7.1 Data Preprocessing Steps

- Missing Values: Missing values in the dataset are checked and handled.
- Duplicate Values: Duplicate records are checked and removed if present.
- Data Types: Data types of all columns are checked and corrected where required.
- Invalid Values: Incorrect or unrealistic values are identified and handled.
- Outliers: Extreme values in numerical columns are identified.
- Inconsistent Values: Categorical values such as Brand, Product and City are checked for consistency.

7.2 code:

```
# Check missing values
print(data.isnull().sum())
# Check duplicate values
print("Duplicate rows:", data.duplicated().sum())
# Convert Date column
data["Date"] = pd.to_datetime(data["Date"])
# Check data types
print(data.dtypes)
# Basic statistics
display(data.describe())
```

Output:

```
Invalid Quantity: 0
Invalid Revenue: 0

Missing Values After Cleaning:
Order_ID      0
Date          0
Brand         0
Product       0
City          0
Customer_Type 0
AB_Group      0
Advertising_Type 0
Price         0
Quantity      0
Discount_Percent 0
Revenue       0
Marketing_Spend 0
Rating        0
Demand        0
Converted     0
Sales_Units   0
dtype: int64

Final Dataset Shape:
(180, 17)
```

8. Exploratory Data Analysis

8.1 Basic Dataset Information

EDA is used to understand the Branded Electronics Sales dataset before performing further analysis.

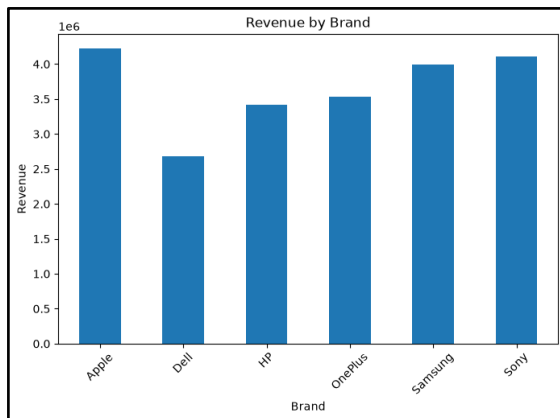
The following points are checked:

- First 5 records of the dataset
- Last 5 records of the dataset
- Number of rows and columns
- Column names
- Data types
- Missing values
- Duplicate records
- Unique values
- Basic statistical summary

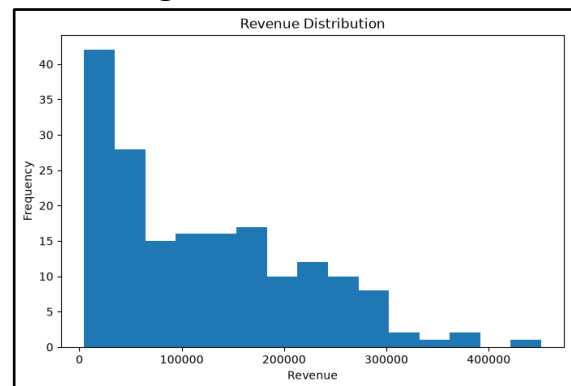
8.2 Data Visualization

The following charts are used for analysis:

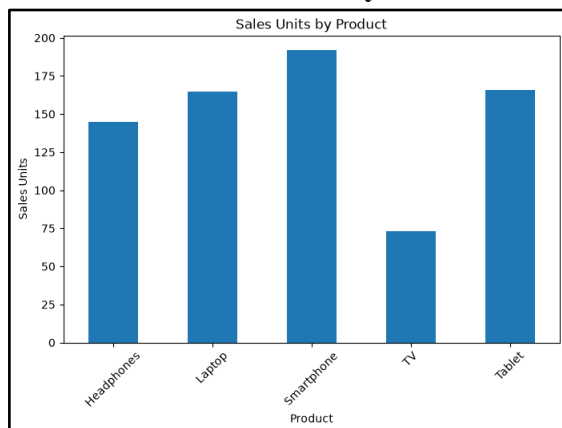
- **Bar Chart – Revenue by Brand**



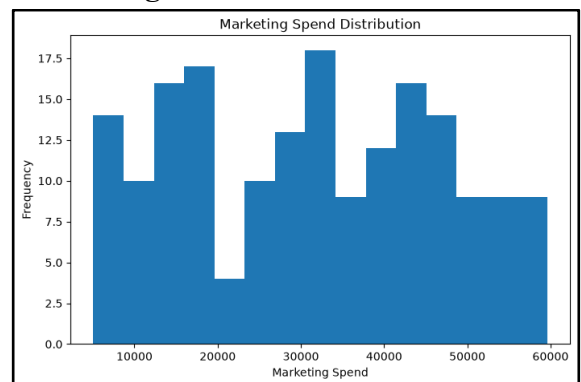
- **Histogram – Price**



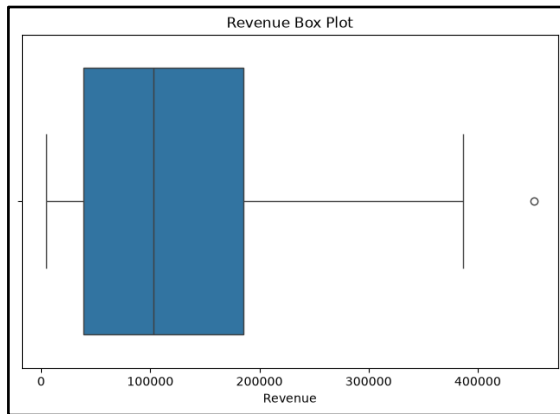
- **Bar Chart – Sales units by Product**



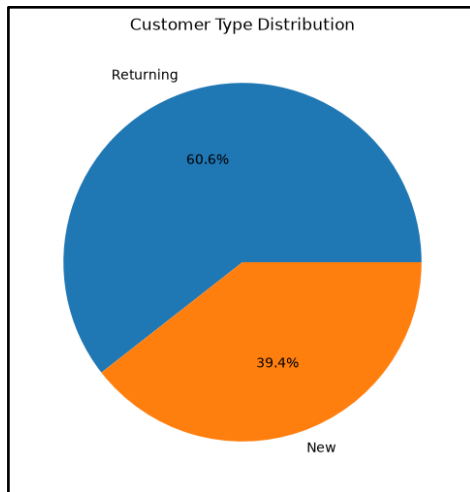
- **Histogram – Revenue**



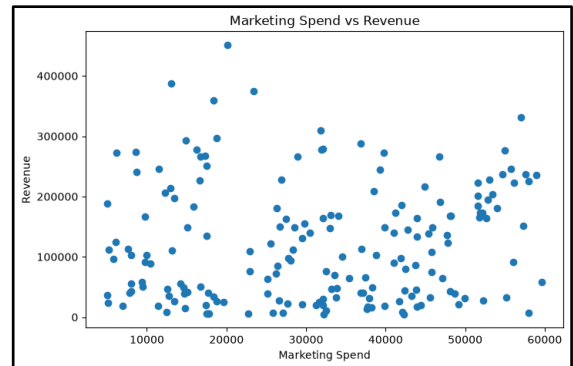
- **Box Plot – Revenue**



- **Pie Chart – Customer Type**



- **Scatter Plot – Marketing Spend vs Revenue**



9. Descriptive Statistical Analysis

Descriptive statistics are used to summarize the main characteristics of the Branded Electronics Sales dataset.

The following measures are calculated:

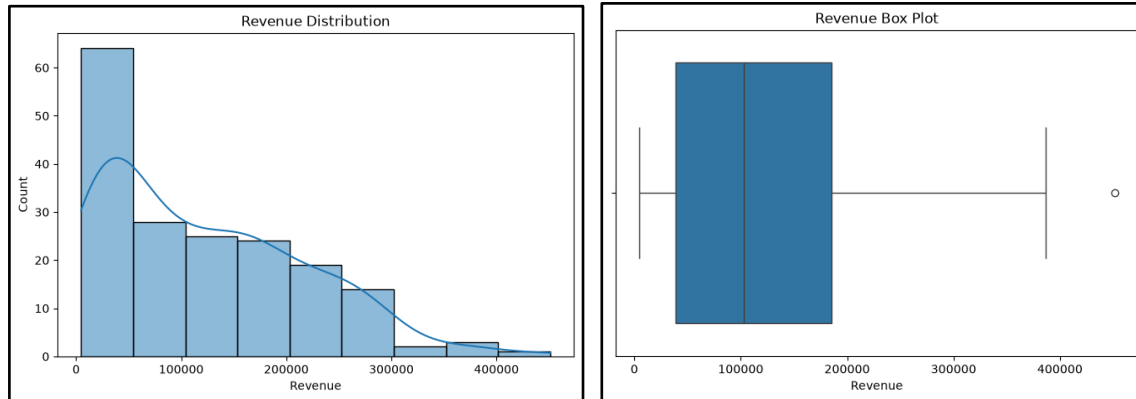
- Mean – Average value
- Median – Middle value
- Mode – Most frequently occurring value
- Minimum – Smallest value
- Maximum – Largest value
- Range – Difference between maximum and minimum
- Variance – Measures data variability
- Standard Deviation – Measures the spread of data

These measures help to understand the sales, price, quantity, discount, revenue, marketing spend, rating and demand of the business.

10. Distribution Analysis

Distribution analysis is used to understand how the numerical values are spread in the Branded Electronics Sales dataset.

- Histogram: Used to observe the distribution of variables such as Price and Revenue.
- Box Plot: Used to understand the spread of data and identify outliers.
- Mean and Median: Compared to understand the shape of the distribution.
- Skewness: Used to check whether the data is symmetric or skewed.



Observation

The distribution analysis helps to understand the central tendency, spread, shape and outliers in the sales data.

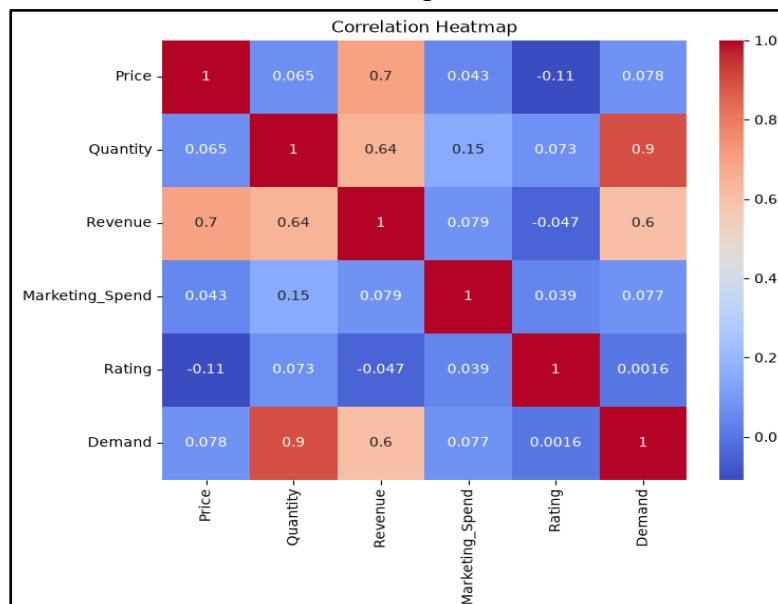
11. Correlation Analysis

Correlation analysis is used to study the relationship between numerical variables in the Branded Electronics Sales dataset.

Three or more numerical variables such as Price, Quantity, Revenue, Marketing Spend and Sales Units are compared.

- Positive Correlation: Both variables increase together.
- Negative Correlation: One variable increases while the other decreases.
- Strong Correlation: Variables have a strong relationship.
- Weak Correlation: Variables have a weak relationship.

A correlation matrix and heatmap are used to visualize these relationships.



Observation

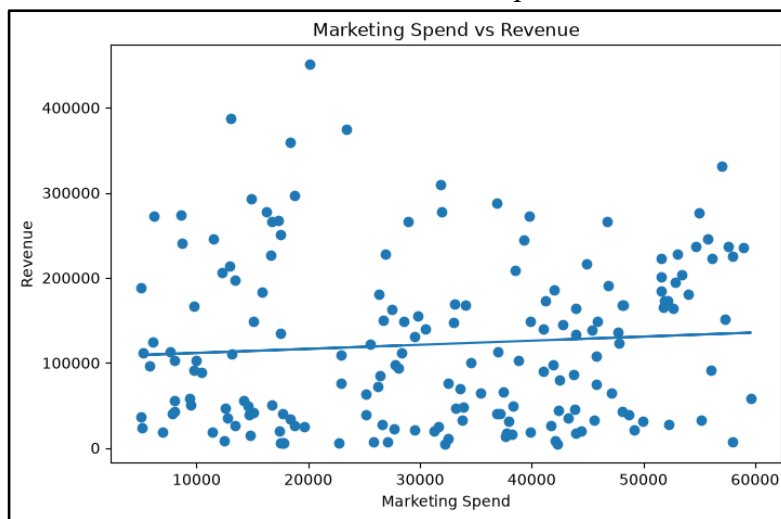
Correlation analysis helps to identify which factors are strongly or weakly related to Revenue and sales performance.

12. Regression Analysis

Regression analysis is used to understand the relationship between two variables and to predict one variable based on another.

For this project:

- Independent Variable: Marketing Spend
- Dependent Variable: Revenue
- A scatter plot with a regression line is used.
- The regression equation is calculated.
- R^2 value is used to measure how well Marketing Spend explains Revenue.
- MSE/RMSE is used to measure prediction error.



Observation

Regression analysis helps to understand whether marketing spending has an effect on revenue and can be used to predict revenue based on marketing expenditure.

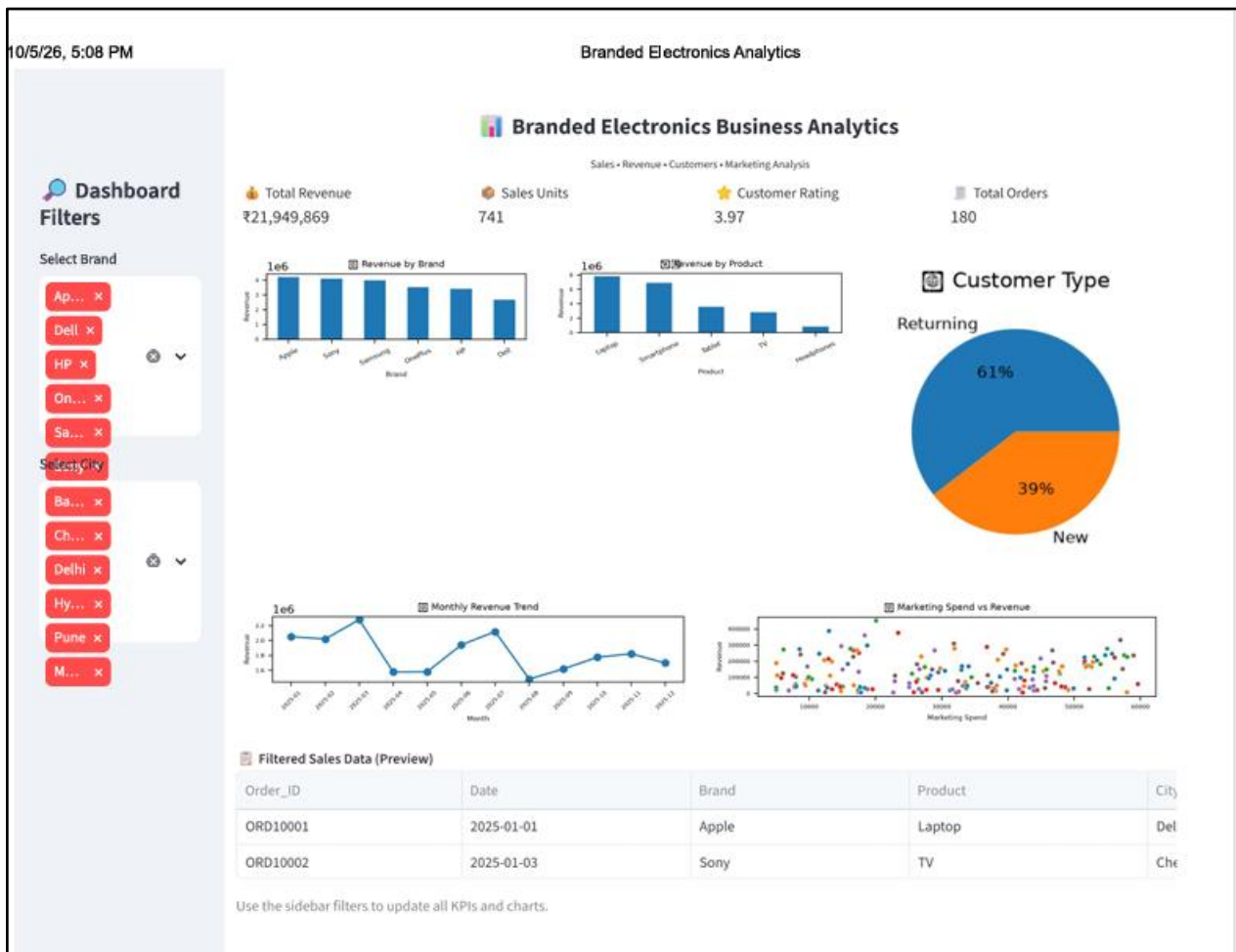
13. Business Intelligence Dashboard

An interactive dashboard was created to analyze the Branded Electronics Sales dataset and understand overall business performance.

The dashboard contains:

- KPI Cards: Total Revenue, Total Sales Units, Average Rating and Marketing Spend.
- Bar Chart: Revenue by Brand.
- Pie Chart: Customer Type Distribution.
- Line Chart: Monthly Revenue Trend.
- Filters: Brand and City filters.
- Titles and Labels: All charts are provided with appropriate titles and labels.

Dashboard Screenshot



13.1 Insights from Dashboard

1. The dashboard helps compare the revenue performance of different brands.
2. Product-wise revenue can be analyzed to identify better-performing products.
3. The monthly revenue chart helps identify sales trends over time.
4. Customer Type analysis shows the distribution of New and Returning customers.
5. Filters allow users to analyze sales for a specific brand or city.

14. Probability and Probability Distributions.

Probability is used to measure the chance of an event occurring in the Branded Electronics business.

A simple probability experiment is performed using variables such as Customer Conversion, Product Sales and Customer Type.

The following probability distributions are also studied:

- Normal Distribution: Used to understand normally distributed numerical data.
- Uniform Distribution: Each possible outcome has an equal chance.
- Exponential Distribution: Used to represent the time between events.

Observation

Probability analysis helps in understanding customer conversion, sales possibilities and business uncertainty, which supports better decision-making.

15. Decision Making Under Uncertainty

Decision-making under uncertainty is used to select the best business option when the future outcome is uncertain.

For this project:

- Different business alternatives are considered.
- Possible outcomes and their probabilities are identified.
- Expected value is calculated for each alternative.
- A decision tree is used to represent the choices and outcomes.
- The alternative with the better expected value is selected.

Observation

This analysis helps the electronics business to make better decisions about marketing and sales strategies under uncertain conditions.

16. Sampling and Sampling Distribution

Sampling is used to select a smaller group of data from the complete Branded Electronics dataset for analysis.

- Population: Complete dataset of 180 records.
- Sample: A selected portion of the dataset.
- Sample Size: Different sample sizes such as 20, 30 and 50 are used.
- Sampling Technique: Random sampling is used.
- Multiple random samples are taken and their sample means are calculated.
- A histogram is used to show the distribution of sample means.
- The Central Limit Theorem (CLT) is studied using the sample means.

Observation

Sampling helps to understand the overall business data using a smaller representative group, saving time and effort while maintaining useful statistical information.

17. Confidence Interval Estimation

Confidence interval is used to estimate the range in which the true population value is likely to lie.

For this project, confidence intervals are calculated at:

- 90% Confidence Level
- 95% Confidence Level
- 99% Confidence Level

The 95% confidence interval is mainly used for business interpretation.

Observation

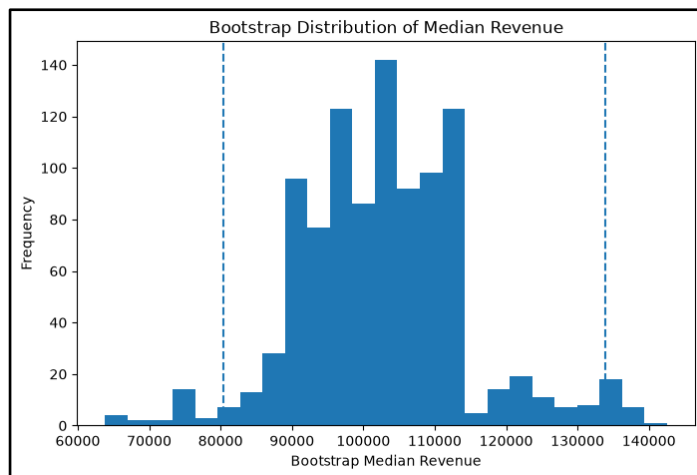
Confidence intervals help the business understand the possible range of values for important variables and make decisions with greater confidence.

18. Bootstrapping

Bootstrapping is a statistical technique used to estimate the uncertainty of a sample statistic by repeatedly creating new samples from the original data.

For this project:

- A numerical variable is selected.
- Samples are created with replacement.
- The process is repeated 1000 times.
- The median is calculated for each sample.
- A bootstrap distribution is created.
- A 95% confidence interval is calculated.



Observation

Bootstrapping helps to understand the reliability and variation of the selected business statistic without requiring many new observations.

19. Hypothesis Testing

Hypothesis testing is used to test a business assumption using statistical evidence.

For this project:

- Null Hypothesis (H_0): There is no significant difference or effect.
- Alternative Hypothesis (H_1): There is a significant difference or effect.
- A suitable statistical test is applied.
- Significance level (α) = 0.05 is used.
- The test statistic and p-value are calculated.
- Based on the p-value, H_0 is accepted or rejected.

Observation

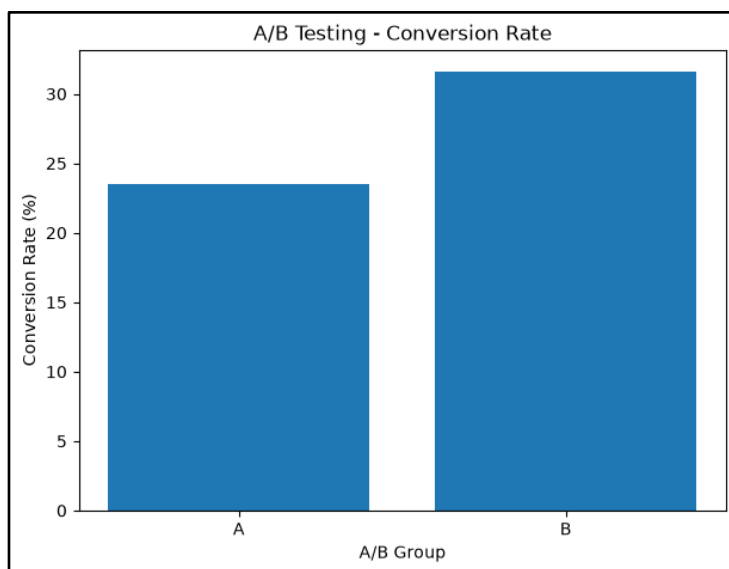
Hypothesis testing helps to determine whether an observed result in the branded electronics sales data is statistically significant and supports business decision-making.

20. A/B Testing

A/B testing is used to compare two different versions or strategies and identify which performs better.

For this project:

- Group A: One marketing strategy
- Group B: Another marketing strategy
- Customer conversion rates of both groups are compared.
- Null Hypothesis (H_0): There is no significant difference between Group A and Group B.
- Alternative Hypothesis (H_1): There is a significant difference between Group A and Group B.
- A statistical test is used to compare the conversion rates.



Observation

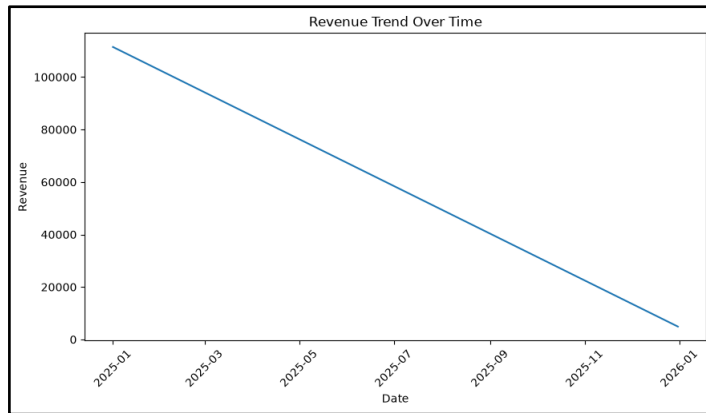
A/B testing helps the business identify which marketing strategy gives better customer conversion and supports data-driven marketing decisions.

21. Time Series Analysis

Time series analysis is used to study changes in sales revenue over time.

For this project:

- The Date column is used as the time variable.
- Revenue is analyzed for different time periods.
- A line chart is used to identify trends.
- Exponential Smoothing is used for forecasting.
- ARIMA is also applied to predict future revenue.
- Forecasting results are compared using MAE, MSE and RMSE.



Observation

Time series forecasting helps the business to understand sales trends and predict future revenue, which supports better planning and decision-making.

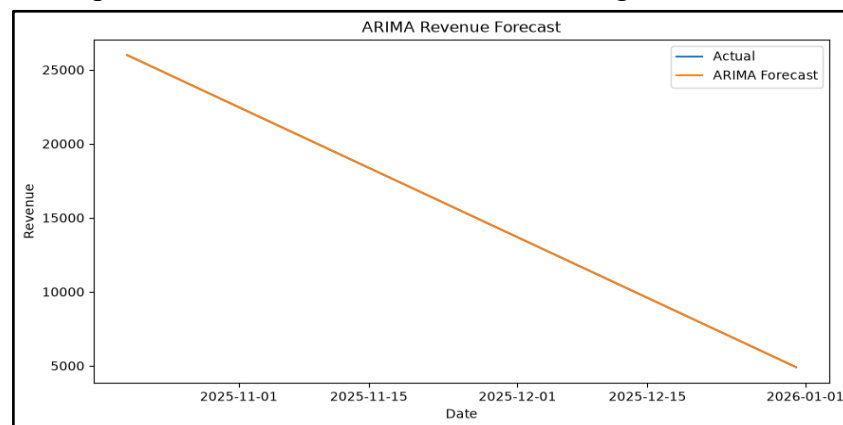
22. Forecast Accuracy

Forecast accuracy is used to measure how close the predicted revenue is to the actual revenue.

The following measures are used:

- MAE (Mean Absolute Error): Measures the average prediction error.
- MSE (Mean Squared Error): Measures the squared prediction error.
- RMSE (Root Mean Squared Error): Measures the average prediction error in the original units.

A comparison table is created for the forecasting models.



Observation

Forecast accuracy helps to identify which forecasting model gives better revenue predictions for the branded electronics business.

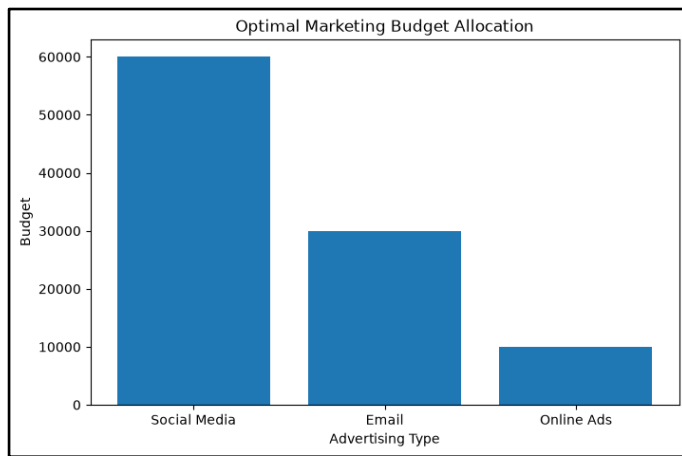
23. Optimization Modeling

Optimization is used to find the best possible business decision while considering available resources and limitations.

For this project:

- Business decision variables are identified.
- A profit objective is defined.
- Budget and other business constraints are considered.
- The optimization problem is solved using Python.

- The best solution is identified based on the objective.



Observation

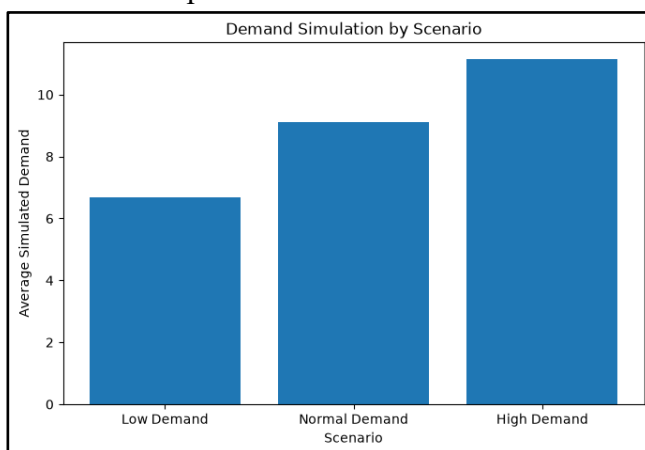
Optimization helps the business decide how to use its available budget and resources efficiently to achieve better business performance.

24. Simulation Modeling

Simulation is used to imitate real-life business situations and understand their possible outcomes.

For this project:

- Different business scenarios are created.
- Random values are generated for the simulation.
- At least 3 scenarios are compared.
- Business performance is measured for each scenario.
- A comparison chart is used to show the results.



Observation

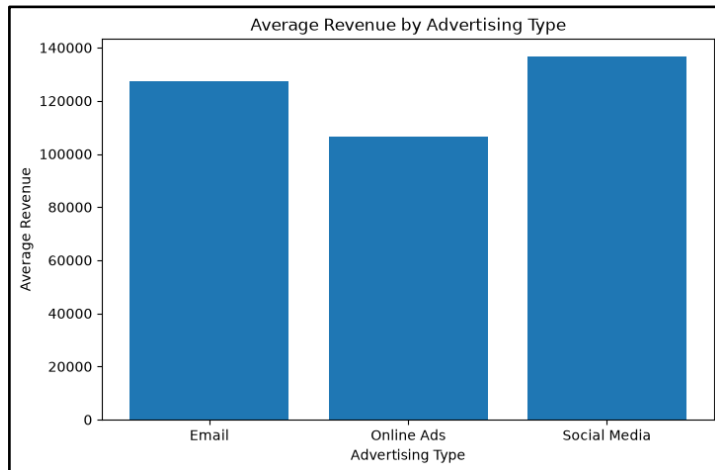
Simulation helps the business understand possible outcomes under different conditions and choose a suitable strategy for better performance.

25. ANOVA

ANOVA is used to compare the average revenue/sales of three or more groups.

For this project:

- Different groups such as Advertising Type are compared.
- H_0 : All group means are equal.
- H_1 : At least one group mean is different.
- One-way ANOVA is applied.
- The F-statistic and p-value are calculated.
- Significance level $\alpha = 0.05$ is used.



Observation

ANOVA helps to determine whether there is a significant difference in business performance among different groups.

26. Experimental Design

Experimental design is used to study how different factors affect business results.

For this project:

- Factors: Advertising Type and Discount Percentage
- Levels: Different advertising types and discount levels
- Response Variable: Revenue / Sales
- Experimental Units: Customer orders
- The effect of different factors on sales is analyzed.

Observation

Experimental design helps the business understand which factors influence sales performance and supports better marketing decisions.

27. Factorial Experiment

Factorial design is used to study the combined effect of two or more factors on business performance.

For this project:

- Factor 1: Advertising Type
- Factor 2: Discount Percentage
- Response Variable: Revenue / Sales
- Main effects of both factors are analyzed.
- The interaction effect between Advertising Type and Discount is also studied.

Observation

Factorial analysis helps the business understand which combination of advertising and discount strategy can improve sales performance.

28. Business Insights and Findings

The analysis of the Branded Electronics Sales dataset provides the following key insights:

- Sales performance varies across different brands and products.
- Revenue is affected by factors such as price, quantity and sales units.
- Customer conversion can be compared between A/B groups.
- Different advertising methods show different sales performance.
- Marketing spending can be analyzed to understand its relationship with revenue.
- Time series analysis helps identify revenue trends over time.
- Statistical analysis helps identify significant differences between business groups.
- Forecasting helps estimate future revenue.
- Optimization helps in making better use of the available budget and resources.
- Simulation helps compare different business scenarios.

29. Business Recommendations

Based on the analysis of the Branded Electronics Sales dataset, the following recommendations can be made:

- Focus on high-performing brands and products.
- Improve marketing strategies based on A/B testing results.
- Use suitable advertising methods to improve sales.
- Plan discounts carefully to maintain better revenue.
- Use sales forecasting for future business planning.
- Allocate the marketing budget using optimization techniques.
- Use customer and sales data for better decision-making.

30. Conclusion

The Branded Electronics Sales Business Analytics project uses different statistical and analytical techniques to understand business performance.

The analysis covers data preprocessing, EDA, descriptive statistics, correlation, regression, probability, sampling, hypothesis testing, A/B testing, forecasting, optimization, simulation and ANOVA.

Overall, the project helps in understanding sales patterns, customer behaviour and business performance and supports data-driven decision-making.

31. Appendix

The appendix contains the supporting materials used in the project:

- Complete Python code
- Dataset ([Branded_Electronics_Business_Analytics_Project_Dataset.csv](#))
- EDA charts and graphs
- Statistical analysis outputs
- Regression and forecasting results
- Decision tree
- Optimization results
- Simulation results
- ANOVA and factorial analysis
- Streamlit dashboard screenshots